Number & Algebra

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ANSWER KEY — COMPLETE WORKED SOLUTIONS

Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 **[Foundational]** • **Target: Solve by factorisation**

Consider the quadratic function $f(x) = x^2 + bx + c$. The graph of $f(x)$ is shown below. The vertex of the parabola is at $(2, -5)$ and it passes through the point $(0, -1)$. Determine the values of b and c.

The vertex form of a quadratic function is $f(x) = a(x - h)^2 + k$, where (h, k) is the vertex. Given the vertex is $(2, -5)$, we have $f(x) = a(x - 2)^2 - 5$. Since the parabola passes through $(0, -1)$, we can substitute these values into the equation: $-1 = a(0 - 2)^2 - 5$ $-1 = a(4) - 5$ $4 = 4a$ $a = 1$. So, the function in vertex form is $f(x) = 1(x - 2)^2 - 5$. Expanding this, we get: $f(x) = (x^2 - 4x + 4) - 5$ $f(x) = x^2 - 4x - 1$. Comparing this to the given form $f(x) = x^2 + bx + c$, we can identify the coef.

2 **[Foundational]** • **Target: Solve quadratics by completing the square**

The graph of the quadratic function $f(x) = 2x^2 - kx + 5$ passes through the point (3, 14). Find the value of k.

The graph of the function $f(x) = 2x^2 - kx + 5$ passes through the point (3, 14). This means that when $x = 3$, $f(x) = 14$. We can substitute these values into the function's equation: $14 = 2(3)^2 - k(3) + 5$ $14 = 2(9) - 3k + 5$ $14 = 18 - 3k + 5$ $14 = 23 - 3k$ Now, we need to solve for k: $3k = 23 - 14$ $3k = 9$ $k = \frac{9}{3}$ $k = 3$

3 **[Foundational]** • **Target: Solve using the quadratic formula**

The diagram shows the graph of the function $f(x) = ax^2 + bx + c$. The graph passes through the points $(1, -4)$, $(2, -5)$, and $(4, -1)$. Determine the value of a.

Since the graph of $f(x) = ax^2 + bx + c$ passes through the given points, we can form a system of linear equations by substituting the coordinates of these points into the function's equation: 1. For $(1, -4)$: $a(1)^2 + b(1) + c = -4 \Rightarrow a + b + c = -4$ (Equation 1) 2. For $(2, -5)$: $a(2)^2 + b(2) + c = -5 \Rightarrow 4a + 2b + c = -5$ (Equation 2) 3. For $(4, -1)$: $a(4)^2 + b(4) + c = -1 \Rightarrow 16a + 4b + c = -1$ (Equation 3) Now, we solve this system of equations. Subtract Equation 1 from Equatio.

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4 [Foundational] • Target: Discriminant of a quadratic

Consider the two functions $f(x) = \log_2(x)$ and $g(x) = 3 - x$. The graphs of $f(x)$ and $g(x)$ intersect at a point P, and $f(x)$ intersects the x-axis at point Q. A line segment is drawn from Q to a point R on the graph of $g(x)$ such that QR is perpendicular to the x-axis. Find the coordinates of the point R.

The graph of $f(x) = \log_2(x)$ intersects the x-axis when $f(x) = 0$. Thus, $\log_2(x) = 0$, which implies $x = 2^0 = 1$. So, point Q has coordinates (1, 0). The line segment QR is perpendicular to the x-axis, meaning that the x-coordinate of R is the same as the x-coordinate of Q. Therefore, the x-coordinate of R is 1. Point R lies on the graph of $g(x) = 3 - x$. Substituting the x-coordinate of R into the equation for $g(x)$, we get $g(1) = 3 - 1 = 2$. Thus, the coordinates of point R are (1, 2). Let's check the.

5 [Foundational] • Target: Solve quadratics using various methods

Consider the function $f(x) = -x^2 + 6x - 5$. The graph of this function is shown below. A tangent line to the graph at point $P(2, 3)$ intersects the x-axis at point Q. Determine the coordinates of point Q.

To find the equation of the tangent line, we first need the derivative of the function $f(x) = -x^2 + 6x - 5$. The derivative is $f'(x) = -2x + 6$. The slope of the tangent line at point $P(2, 3)$ is $f'(2) = -2(2) + 6 = -4 + 6 = 2$. Using the point-slope form of a line, $y - y_1 = m(x - x_1)$, with $(x_1, y_1) = (2, 3)$ and $m = 2$, the equation of the tangent line is $y - 3 = 2(x - 2)$. This simplifies to $y - 3 = 2x - 4$, or $y = 2x - 1$. To find the x-intercept (point Q), we set $y = 0$: $0 = 2x - 1$. Solving.

6 [Foundational] • Target: Graph basic quadratics

Consider the function $f(x) = x^2 - 4x + 3$ and the line $g(x) = x - 1$. The graph of these functions is shown below. Find the coordinates of the point where the line intersects the parabola.

To find the intersection points, we set the two functions equal to each other: $f(x) = g(x)$ $x^2 - 4x + 3 = x - 1$ $x^2 - 5x + 4 = 0$ Factoring the quadratic equation: $(x - 1)(x - 4) = 0$ This gives us two possible x-values for the intersection points: $x = 1$ or $x = 4$. Using $g(x) = x - 1$ to find the corresponding y-values: For $x = 1$: $y = 1 - 1 = 0$. The intersection point is $(1, 0)$. For $x = 4$: $y = 4 - 1 = 3$. The intersection point is $(4, 3)$. Looking at the graph, one of the intersection points is $(1, 0)$.

7 [Foundational] • Target: Solve by factorisation

Consider the function $f(x) = x^2 - 2x + 1$. Determine the y-coordinate of the vertex of the graph of $y = f(x)$.

The graph of a quadratic function $f(x) = ax^2 + bx + c$ is a parabola. The x-coordinate of the vertex is given by $x = -\frac{b}{2a}$. For the function $f(x) = x^2 - 2x + 1$, we have $a = 1$ and $b = -2$. Therefore, the x-coordinate of the vertex is $x = -\frac{-2}{2(1)} = \frac{2}{2} = 1$. To find the y-coordinate of the vertex, we substitute this value of x back into the function: $f(1) = (1)^2 - 2(1) + 1 = 1 - 2 + 1 = 0$. Thus, the y-coordinate of the vertex is 0.

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Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 **[Intermediate]** • **Target: Solve quadratics by completing the square**

Consider the function $f(x) = -x^2 + 6x - 5$. Determine the coordinates of the vertex of the graph of $y = f(x)$.

The given function is a quadratic of the form $f(x) = ax^2 + bx + c$, where $a = -1$, $b = 6$, and $c = -5$. The x-coordinate of the vertex of a parabola is given by $x = -\frac{b}{2a}$. Substituting the values, we get $x = -\frac{6}{2(-1)} = -\frac{6}{-2} = 3$. To find the y-coordinate of the vertex, substitute $x = 3$ into the function: $f(3) = -(3)^2 + 6(3) - 5 = -9 + 18 - 5 = 9 - 5 = 4$. Therefore, the coordinates of the vertex are (3, 4).

9 **[Intermediate]** • **Target: Solve using the quadratic formula**

The graph of the function $f(x) = x^2 - 6x + k$ is shown below. The graph passes through the point $(5, -2)$. Find the value of k.

The function is given by $f(x) = x^2 - 6x + k$. We are given that the graph of this function passes through the point $(5, -2)$. This means that when $x = 5$, $f(x) = -2$. Substituting these values into the function, we get: $-2 = (5)^2 - 6(5) + k$ $-2 = 25 - 30 + k$ $-2 = -5 + k$ To solve for k, we add 5 to both sides of the equation: $-2 + 5 = k$ $3 = k$ Therefore, the value of k is 3.

10 **[Intermediate]** • **Target: Discriminant of a quadratic**

Consider the function $f(x) = e^{-x^2}$ and the line $y = k$. The graph of $y = f(x)$ is concave up for $x \in (-\infty, -a) \cup (a, \infty)$ and concave down for $x \in (-a, a)$. Find the exact value of a^2.

To determine the intervals of concavity, we need to find the second derivative of $f(x) = e^{-x^2}$. First derivative: $f'(x) = e^{-x^2} \cdot (-2x) = -2xe^{-x^2}$. Second derivative using the product rule: $f''(x) = (-2)e^{-x^2} + (-2x) \cdot e^{-x^2} \cdot (-2x)$ $f''(x) = -2e^{-x^2} + 4x^2e^{-x^2} = e^{-x^2}(4x^2 - 2)$. The concavity of $f(x)$ is determined by the sign of $f''(x)$. Since e^{-x^2} is always positive, the sign of $f''(x)$ is determined by the sign of $4x^2 - 2$. The function is concave down when.

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11 [Intermediate] • Target: Solve quadratics using various methods

The depth of water $h(t)$ in meters at a harbor entrance t hours after midnight is modelled by $h(t) = 3.5\cos\left(\frac{\pi t}{6}\right) + 6.5$ for $0 \le t \le 24$. (a) State the maximum and minimum depths of water and their times of occurrence. (b) Find the period of the tidal oscillation. (c) A cargo ship requires a minimum water depth of 8.25 meters to safely enter the harbor. Find the total number of hours during the first 12 hours of the day when the ship can safely enter.

(a) Maximum depth occurs when $\cos = 1$: $h_{\max} = 6.5 + 3.5 = 10.0$ m. This occurs at $t = 0$, $t = 12$, and $t = 24$ hours. Minimum depth occurs when $\cos = -1$: $h_{\min} = 6.5 - 3.5 = 3.0$ m. This occurs at $t = 6$ and $t = 18$ hours. (b) Period $T = \frac{2\pi}{B} = \frac{2\pi}{\pi/6} = 12$ hours. (c) Solve $h(t) \ge 8.25$ over $[0, 12]$.

12 [Intermediate] • Target: Graph basic quadratics

Consider the function $f(x) = x^3 - 6x^2 + 5$. Find the exact value of the sum of the x-coordinates of the local maximum and local minimum points of the graph of $y = f(x)$.

To find the local maximum and minimum points, we first need to find the critical points by setting the first derivative of $f(x)$ to zero. First derivative: $f'(x) = \frac{d}{dx}(x^3 - 6x^2 + 5) = 3x^2 - 12x$. Set $f'(x) = 0$ to find critical points: $3x^2 - 12x = 0$ $3x(x - 4) = 0$ So, the critical points are $x = 0$ and $x = 4$. To determine whether these are local maxima or minima, we can use the second derivative test. Second derivative: $f''(x) = \frac{d}{dx}(3x^2 - 12x) = 6x - 12$. Evaluate $f''(x)$ at the cr.

13 [Intermediate] • Target: Solve by factorisation

The function $f(x) = \frac{2x-5}{x+3}$ is defined for $x \neq -3$. (a) Find the coordinates of the axes intercepts of the graph of $y = f(x)$. (b) Write down the equations of the vertical and horizontal asymptotes. (c) Find an expression for $f^{-1}(x)$ and state its domain and range.

(a) y-intercept (at $x = 0$): $f(0) = \frac{-5}{3} = -\frac{5}{3}$, giving $\left(0, -\frac{5}{3}\right)$. x-intercept (at $y = 0$): $2x - 5 = 0 \Rightarrow x = 2.5$, giving $(2.5, 0)$. (b) Vertical asymptote where denominator is zero: $x = -3$. Horizontal asymptote: $y = \lim_{x \to \pm\infty} \frac{2x-5}{x+3} = 2$. (c) Set $y = \frac{2x-5}{x+3} \Rightarrow y(x + 3) = 2x - 5 \Rightarrow xy + 3y = 2x - 5$. Rearrange.

14 [Intermediate] • Target: Solve quadratics by completing the square

In the triangle ABC shown, point D lies on AB and point E lies on AC such that DE is parallel to BC. Given that AD = 6, DB = 4, and the area of triangle ADE is 27 square units, find the area of triangle ABC.

Since DE is parallel to BC, by the property of similar triangles, triangle ADE is similar to triangle ABC. The ratio of corresponding sides is $\frac{AD}{AB} = \frac{AD}{AD + DB} = \frac{6}{6+4} = \frac{6}{10} = \frac{3}{5}$. For similar triangles, the ratio of their areas is the square of the ratio of their corresponding sides. Therefore, $\frac{\text{Area}(ADE)}{\text{Area}(ABC)} = \left(\frac{AD}{AB}\right)^2 = \left(\frac{3}{5}\right)^2 = \frac{9}{25}$. We are given that the area of triangle ADE is 27 square.

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Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

15 [Challenging] • Target: Solve using the quadratic formula • Cross-Lesson: 6.01 AI SL — Specific Topics

The box plot below shows the distribution of scores for a class on a recent statistics quiz. The five-number summary is given by minimum, first quartile (Q1), median, third quartile (Q3), and maximum. If the interquartile range (IQR) is 25, and the median score is 70, find the value of Q1.

The interquartile range (IQR) is defined as the difference between the third quartile (Q3) and the first quartile (Q1), i.e., $\text{IQR} = Q_3 - Q_1$. We are given that the IQR is 25. The median of the dataset is given as 70. In a box plot, the median is represented by the line inside the box. Therefore, $Q_2 = 70$. The box plot also indicates that the third quartile (Q3) is 70. This is because the right edge of the box aligns with the median line, indicating $Q_3 = 70$. Now we can use the IQR formula to find Q1.

16 [Challenging] • Target: Discriminant of a quadratic • Cross-Lesson: 2.01 Functions

Evaluate the definite integral $\int_0^{\sqrt{7}} \frac{4x}{\sqrt{x^2+9}}\,dx$ using the substitution $u = x^2 + 9$.

1. Let $u = x^2 + 9$. Then $du = 2x\,dx \Rightarrow 4x\,dx = 2\,du$. 2. Transform the limits of integration: When $x = 0$: $u = 0^2 + 9 = 9$. When $x = \sqrt{7}$: $u = (\sqrt{7})^2 + 9 = 7 + 9 = 16$. 3. Rewrite the integral in terms of u: $\int_9^{16} \frac{2}{\sqrt{u}}\,du = 2\int_9^{16} u^{-1/2}\,du = 2[2u^{1/2}]_9^{16} = 4[\sqrt{u}]_9^{16}$. 4. Evaluate at the boundaries: $4(\sqrt{16} - \sqrt{9}) = 4(4 - 3) = 4(1) = 4$. The exact value of the integral is 4.

17 [Challenging] • Target: Solve quadratics using various methods • Cross-Lesson: 3.01 Geometry & Trigonometry

Solve the trigonometric equation $2\sin^2\theta + 3\cos\theta - 3 = 0$ for $0 \le \theta \le 2\pi$, giving exact radian answers.

1. Substitute $\sin^2\theta = 1 - \cos^2\theta$: $2(1 - \cos^2\theta) + 3\cos\theta - 3 = 0 \Rightarrow 2 - 2\cos^2\theta + 3\cos\theta - 3 = 0$. $-2\cos^2\theta + 3\cos\theta - 1 = 0 \Rightarrow 2\cos^2\theta - 3\cos\theta + 1 = 0$. 2. Factor the quadratic in $\cos\theta$: $(2\cos\theta - 1)(\cos\theta - 1) = 0$. 3. Solve each branch: Branch 1: $2\cos\theta - 1 = 0 \Rightarrow \cos\theta = \frac{1}{2}$. Reference angle: $\alpha = \frac{\pi}{3}$. In $[0, 2\pi]$, cosine is positive in Q1 and Q4.

18 [Challenging] • Target: Graph basic quadratics • Cross-Lesson: 4.01 Statistics & Probability

Consider the function $f(x) = x^2 - 4x + 3$ and the function $g(x) = x - 1$. The graphs of $y = f(x)$ and $y = g(x)$ intersect at points A and B. One of these intersection points is $(1, 0)$. Find the coordinates of the other intersection point.

To find the intersection points of the two graphs, we set $f(x) = g(x)$. $x^2 - 4x + 3 = x - 1$ Subtract x from both sides: $x^2 - 5x + 3 = -1$ Add 1 to both sides: $x^2 - 5x + 4 = 0$ Factor the quadratic equation: $(x - 1)(x - 4) = 0$ This gives us two possible x-coordinates for the intersection points: $x = 1$ and $x = 4$. We are given that one intersection point has an x-coordinate of 1. To find the corresponding y-coordinate, we can substitute $x = 1$ into either $f(x)$ or $g(x)$. Using $g(x) = x - 1$, we get.

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19 [Challenging] • Target: Solve by factorisation • Cross-Lesson: 5.01 Calculus

Solve the logarithmic equation $\log_3(x+5)+\log_3(x-1)=2+\log_3(2)$ for all real values of x.

1. Domain constraints: $x+5>0 \Rightarrow x>-5$ and $x-1>0 \Rightarrow x>1$. Thus $x>1$. 2. Rewrite the right-hand side using $2=\log_3(3^2)=\log_3(9)$: $2+\log_3(2)=\log_3(9)+\log_3(2)=\log_3(18)$. 3. Combine the left-hand side: $\log_3((x+5)(x-1))=\log_3(18) \Rightarrow (x+5)(x-1)=18$. 4. Expand and solve the quadratic equation: $x^2+4x-5=18 \Rightarrow x^2+4x-23=0$. 5. Quadratic formula.

20 [Challenging] • Target: Solve quadratics by completing the square • Cross-Lesson: 6.01 AI SL — Specific Topics

A discrete random variable X follows a binomial distribution $X \sim B(15, 0.35)$. (a) Find $P(X=5)$. (b) Find $P(X \geq 4)$. (c) Calculate the mean $E(X)$ and standard deviation σ of X.

(a) $P(X=5)=\binom{15}{5}(0.35)^5(0.65)^{10}=3003\times(0.005252)\times(0.013463)\approx 0.2123$. (b) $P(X\geq 4)=1-P(X\leq 3)$. Using GDC `binomcdf(15, 0.35, 3)`: $P(X\leq 3)\approx 0.1727$. $P(X\geq 4)=1-0.1727=0.8273$ (82.7%). (c) Mean: $E(X)=np=15\times 0.35=5.25$. Variance: $\text{Var}(X)=np(1-p)=15\times 0.35\times 0.65=3.4125$. Standard deviation: $\sigma=\sqrt{3.4125}\approx 1.847\approx 1.85$.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

21 [Foundational] • Target: Solve by factorisation

For the model $f(x)=2x+3$ in solve by factorisation, evaluate $f(2)$.

a. 7 ✓ **b.** 9

c. 4 **d.** 11

Substitute $x=2$: $f(2)=2(2)+3=7$.

22 [Foundational] • Target: Solve by factorisation

Find the positive solutions to the quadratic equation $x^2-6x+8=0$.

a. $x=2$ and $x=5$ **b.** $x=2$ **and** $x=4$ ✓

c. $x=-2$ and $x=-4$ **d.** $x=6$ and $x=8$

Factoring $(x-2)(x-4)=0$ yields the roots $x=2$ and $x=4$.

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23 [Foundational] • Target: Solve by factorisation

For the model $f(x) = 2x + 3$ in solve by factorisation, evaluate $f(5)$.

a. 13 ✓

b. 15

c. 10

d. 17

Substitute $x = 5$: $f(5) = 2(5) + 3 = 13$.

24 [Foundational] • Target: Solve quadratics by completing the square

For the model $f(x) = 3x + 4$ in solve quadratics by completing the square, evaluate $f(2)$.

a. 10 ✓

b. 13

c. 6

d. 14

Substitute $x = 2$: $f(2) = 3(2) + 4 = 10$.

25 [Foundational] • Target: Solve quadratics by completing the square

Find the positive solutions to the quadratic equation $x^2 - 7x + 10 = 0$.

a. $x = 2$ and $x = 6$

b. $x = 2$ **and** $x = 5$ ✓

c. $x = -2$ and $x = -5$

d. $x = 7$ and $x = 10$

Factoring $(x - 2)(x - 5) = 0$ yields the roots $x = 2$ and $x = 5$.

26 [Foundational] • Target: Solve quadratics by completing the square

For the model $f(x) = 3x + 4$ in solve quadratics by completing the square, evaluate $f(5)$.

a. 19 ✓

b. 22

c. 15

d. 23

Substitute $x = 5$: $f(5) = 3(5) + 4 = 19$.

27 [Foundational] • Target: Solve using the quadratic formula

For the model $f(x) = 4x + 5$ in solve using the quadratic formula, evaluate $f(2)$.

a. 13 ✓

b. 17

c. 8

d. 17

Substitute $x = 2$: $f(2) = 4(2) + 5 = 13$.

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28 [Foundational] • Target: Solve using the quadratic formula

Find the positive solutions to the quadratic equation $x^2 - 8x + 12 = 0$.

a. $x = 2$ and $x = 7$

b. $x = 2$ **and** $x = 6$ ✓

c. $x = -2$ and $x = -6$

d. $x = 8$ and $x = 12$

Factoring $(x - 2)(x - 6) = 0$ yields the roots $x = 2$ and $x = 6$.

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 [Intermediate] • Target: Solve by factorisation

Find the positive solutions to the quadratic equation $x^2 - 4x + 3 = 0$.

a. $x = 1$ and $x = 4$

b. $x = 1$ **and** $x = 3$ ✓

c. $x = -1$ and $x = -3$

d. $x = 4$ and $x = 3$

Factoring $(x - 1)(x - 3) = 0$ yields the roots $x = 1$ and $x = 3$.

30 [Intermediate] • Target: Solve by factorisation

A data set recorded for solve by factorisation consists of the four values $\{10, 14, 18, 24\}$. What is the arithmetic mean?

a. 14.5

b. 15.5

c. 16.5 ✓

d. 18.5

The sum is $10 + 14 + 18 + 24 = 66$, and dividing by 4 gives 16.5.

31 [Intermediate] • Target: Solve by factorisation

For the model $f(x) = 2x + 3$ in solve by factorisation, evaluate $f(4)$.

a. 11 ✓

b. 13

c. 8

d. 15

Substitute $x = 4$: $f(4) = 2(4) + 3 = 11$.

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32 [Intermediate] • Target: Solve by factorisation

Find the positive solutions to the quadratic equation $x^2 - 10x + 24 = 0$.

a. $x = 4$ and $x = 7$

b. $x = 4$ **and** $x = 6$ ✓

c. $x = -4$ and $x = -6$

d. $x = 10$ and $x = 24$

Factoring $(x - 4)(x - 6) = 0$ yields the roots $x = 4$ and $x = 6$.

33 [Intermediate] • Target: Solve quadratics by completing the square

A data set recorded for solve quadratics by completing the square consists of the four values $\{11, 15, 19, 25\}$. What is the arithmetic mean?

a. 15.5

b. 16.5

c. 17.5 ✓

d. 19.5

The sum is $11 + 15 + 19 + 25 = 70$, and dividing by 4 gives 17.5.

34 [Intermediate] • Target: Solve quadratics by completing the square

For the model $f(x) = 3x + 4$ in solve quadratics by completing the square, evaluate $f(3)$.

a. 13 ✓

b. 16

c. 9

d. 17

Substitute $x = 3$: $f(3) = 3(3) + 4 = 13$.

35 [Intermediate] • Target: Solve quadratics by completing the square

Find the positive solutions to the quadratic equation $x^2 - 9x + 18 = 0$.

a. $x = 3$ and $x = 7$

b. $x = 3$ **and** $x = 6$ ✓

c. $x = -3$ and $x = -6$

d. $x = 9$ and $x = 18$

Factoring $(x - 3)(x - 6) = 0$ yields the roots $x = 3$ and $x = 6$.

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36 [Intermediate] • Target: Solve using the quadratic formula

Find the positive solutions to the quadratic equation $x^2 - 6x + 5 = 0$.

a. $x = 1$ and $x = 6$

b. $x = 1$ **and** $x = 5$ ✓

c. $x = -1$ and $x = -5$

d. $x = 6$ and $x = 5$

Factoring $(x - 1)(x - 5) = 0$ yields the roots $x = 1$ and $x = 5$.

37 [Intermediate] • Target: Solve using the quadratic formula

For the model $f(x) = 4x + 5$ in solve using the quadratic formula, evaluate $f(3)$.

a. 17 ✓

b. 21

c. 12

d. 21

Substitute $x = 3$: $f(3) = 4(3) + 5 = 17$.

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 [Challenging] • Target: Solve using the quadratic formula • Cross-Lesson: 4.01 Statistics & Probability

For the model $f(x) = 4x + 5$ in solve using the quadratic formula, evaluate $f(4)$.

a. 21 ✓

b. 25

c. 16

d. 25

Substitute $x = 4$: $f(4) = 4(4) + 5 = 21$.

39 [Challenging] • Target: Solve using the quadratic formula • Cross-Lesson: 5.01 Calculus

Find the positive solutions to the quadratic equation $x^2 - 12x + 32 = 0$.

a. $x = 4$ and $x = 9$

b. $x = 4$ **and** $x = 8$ ✓

c. $x = -4$ and $x = -8$

d. $x = 12$ and $x = 32$

Factoring $(x - 4)(x - 8) = 0$ yields the roots $x = 4$ and $x = 8$.

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40 [Challenging] • Target: Discriminant of a quadratic • Cross-Lesson: 6.01 AI SL — Specific Topics

A data set recorded for discriminant of a quadratic consists of the four values $\{13, 17, 21, 27\}$. What is the arithmetic mean?

a. 17.5

b. 18.5

c. 19.5 ✓

d. 21.5

The sum is $13 + 17 + 21 + 27 = 78$, and dividing by 4 gives 19.5.

41 [Challenging] • Target: Discriminant of a quadratic • Cross-Lesson: 2.01 Functions

For the model $f(x) = 5x + 6$ in discriminant of a quadratic, evaluate $f(4)$.

a. 26 ✓

b. 31

c. 20

d. 30

Substitute $x = 4$: $f(4) = 5(4) + 6 = 26$.

42 [Challenging] • Target: Discriminant of a quadratic • Cross-Lesson: 3.01 Geometry & Trigonometry

Find the positive solutions to the quadratic equation $x^2 - 11x + 24 = 0$.

a. $x = 3$ and $x = 9$

b. $x = 3$ **and** $x = 8$ ✓

c. $x = -3$ and $x = -8$

d. $x = 11$ and $x = 24$

Factoring $(x - 3)(x - 8) = 0$ yields the roots $x = 3$ and $x = 8$.

43 [Challenging] • Target: Solve quadratics using various methods • Cross-Lesson: 4.01 Statistics & Probability

A data set recorded for solve quadratics using various methods consists of the four values $\{14, 18, 22, 28\}$. What is the arithmetic mean?

a. 18.5

b. 19.5

c. 20.5 ✓

d. 22.5

The sum is $14 + 18 + 22 + 28 = 82$, and dividing by 4 gives 20.5.

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44 [Challenging] • Target: Solve quadratics using various methods • Cross-Lesson: 5.01 Calculus

For the model $f(x) = 6x + 7$ in solve quadratics using various methods, evaluate $f(4)$.

a. 31 ✓

b. 37

c. 24

d. 35

Substitute $x = 4$: $f(4) = 6(4) + 7 = 31$.

45 [Challenging] • Target: Graph basic quadratics • Cross-Lesson: 6.01 AI SL — Specific Topics

For the model $f(x) = 2x + 8$ in graph basic quadratics, evaluate $f(3)$.

a. 14 ✓

b. 16

c. 6

d. 18

Substitute $x = 3$: $f(3) = 2(3) + 8 = 14$.

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